

$$f(x) = \begin{cases} \sin(2x) & , x > 0 \\ a + bx & , x \leq 0 \end{cases}$$

Continuity

$$\Rightarrow \text{ } \Rightarrow f(0) = f(0^+)$$

$$\Rightarrow a = 0$$

$$f'(x) = \begin{cases} 2\cos(2x) & , x > 0 \\ b & , x \leq 0 \end{cases}$$

Differentiability

$$f'(0^+) = f'(0)$$

$$\Rightarrow 2 = b$$

$$\Rightarrow a + b = 2$$